Finite-Moment Nonidentifiability by Hidden Root-of-Unity Shells

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July 2026

Abstract

Growing coefficient prefixes do not by themselves identify a spectral cloud. For every prescribed prefix length N, we construct a conjugation-symmetric shell whose power sums vanish through order N but whose next selected moment is nonzero. Appending the shell to any spectral multiset preserves the entire observed prefix while multiplying the regularized determinant by $1 - (\gamma z)^L$ for some $L > N$. Thus arbitrarily long finite agreement is compatible with a new divisor immediately beyond the checked window. The result is scoped: weighted tail bounds, root transport, or contour control can exclude the construction. It shows exactly why the rate-free prefix theorem of RH-287 cannot replace the weighted gluing leaf of RH-288.

1 Hidden shell

Fix $L \geq 2$ and $\gamma > 0$. Let

$$\mathcal{S}_{L,\gamma} = \{\gamma e^{2\pi i j/L} : 0 \leq j < L\}.$$

This multiset is closed under conjugation.

Theorem 1.1 (prefix invisibility)**.** *For every integer $n \geq 1$,*

$$\sum_{\zeta \in \mathcal{S}_{L,\gamma}} \zeta^n = L\gamma^n \mathbf{1}_{L|n}.$$

Consequently, if $L > N$, every power sum through order N vanishes, while the order-L sum equals $L\gamma^L$.

Proof. Factor out γ^n and sum the finite geometric progression $\sum_{j=0}^{L-1} e^{2\pi i j n/L}$. □

Corollary 1.2 (exact hidden divisor)**.** *The genus-one shell factor is*

$$\prod_{\zeta \in \mathcal{S}_{L,\gamma}} (1 - z\zeta)e^{z\zeta} = 1 - (\gamma z)^L.$$

Proof. The linear exponential terms cancel because the first shell moment is zero. The remaining polynomial identity is the factorization of $1 - (\gamma z)^L$ over the Lth roots of unity. □

2 Nonidentifiability consequence

Let C be any finite spectral multiset and set $\widetilde{C} = C \sqcup \mathcal{S}_{L,\gamma}$. If $L > N$, then

$$\sum_{\lambda \in C} \lambda^n = \sum_{\lambda \in \widetilde{C}} \lambda^n, \qquad 1 \leq n \leq N,$$

but the associated canonical products differ by $1 - (\gamma z)^L$. No rule reading only the first N moments can distinguish them.

The same construction can be applied at a moving $N = N_\sigma$. Without a weighted estimate that suppresses $\gamma_\sigma^{L_\sigma} R^{L_\sigma}$, even a diverging prefix does not determine the analytic factor on $|z| \le R$.

Remark 2.1. This counterexample does not show that the physical folded Gaussian spectrum contains hidden shells. It only proves insufficiency of prefix data. A root-ℓ^1, weighted-Fourier, or contour theorem would add information that the construction ignores. Gates A–E remain false/open.